CSE 2nd Semester Syllabus

Computer Science & Engineering • Rajshahi University of Engineering & Technology

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CSE 1200

Competitive Programming Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Warmup: Basic Input/Output and Conditionals, Basic Mathematics and Introduction to Number Theory, Basic Geometry, Introduction to Different Online Judges.

Loop and Array: Loops and Nested Loops, Array Operations and Techniques i.e., Cumulative Sum, Longest Consecutive Elements, Finding Duplicates etc.

Functions and Strings: Function Building, Specialized Functions, Basic String Operations and Techniques, Substrings, Subsequences, Palindrome.

Searching and Sorting: Searching Techniques i.e., Linear Search, Binary Search, etc., Sorting Techniques i.e., Bubble Sort, Selection Sort, etc.

Advanced Topics: Recursion, Backtracking, Matrix Operations etc.

CSE 1201

Data Structure

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Introduction: Concepts and Examples of Elementary Data Objects, Necessity of Structured Data, Types of Data Structure, Ideas on Linear and Nonlinear Data Structure.

Linear Array: Linear Array & its Representation in Memory, Traversing LA, Insertion & Deletion in LA, Multidimensional Array & its Representation in Memory, Algebra of Matrices, Sparse Matrices.

Stack: Stack Representation & Applications; PUSH and POP Operation on Stack. Polish Notation, Reverse Polish Notation; Evaluation of a Postfix Expression; Transforming Infix Expression into Postfix Expression.

Queue: Its Representation, Insertion & Deletion in Queue, Priority Queues, Recursion: [Factorial Function, Fibonacci Sequence, Ackermann Function, Towers of Hanoi].

Linked List: Linked List & its Representation in Memory, Traversing, Searching, Insertion & Deletion Operation on Linked List, Circular List, Header Linked Lists, Two Way Lists.

Complexity Analysis: Algorithm and Flowchart, Asymptotic Notations: Best case, Worst Case, Average Case, Complexity Analysis of Different Algorithms.

Sorting and Searching: Insertion Sort, Selection Sort, Bubble Sort, Quick Sort, Merge Sort, Binary and Linear Search, Hash Function, Collision Resolution.

Tree: Tree Terminology, Representation of Binary Trees in Memory, Traversing Binary Tree, Binary Search Tree, Insertion & Deletion on Binary Search Tree, Balanced Binary Search Tree, AVL tree, Red Black Tree, Insertion & Deletion on Heap, Heap Sort, B Trees, General Tree.

CSE 1202

Data Structure Sessional

Contact Hours/Week: 3 Hours/Week | Credit Hour: 1.50

Sessional on Data Structure: Linear Array, Stack, Queue, Recursion, Linked List, Sorting & Searching, Tree.

CSE 1203

Object Oriented Programming

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Fundamentals of OOP: Introduction to Object Oriented Programming, Principles of Object-Oriented Design, Encapsulation and Information-hiding, Data Binding, Static and Dynamic Binding.

Classes and Objects: Structure of Class, Access Modifiers, Nested Classes, Abstract Classes, Arrays of Objects, Pointer to Objects, Friend Function, Data Abstraction, Static Variable and Function.

Constructors and Destructors: Default Constructor, Copy Constructor, Dynamic Constructor, Deep Copy and Shallow Copy, Constructor Function for Derived Class and their Order of Execution, Destructor.

Inheritance: Single Inheritance vs. Multiple Inheritance, Mode of Inheritance, Virtual Inheritance, Interface.

Polymorphism: Operator and Function Overloading, Run-Time and Compile Time Polymorphism, Virtual Function, Errors and Exception Handling.

Advanced Topics: Persistent Objects, Objects and Portable Data, UML Basics, Namespace, Package, Templates, JFrame, Multithreading, Concept of MVC Framework.

CSE 1204

Object Oriented Programming Sessional

Contact Hours/Week: 3 Hours/Week | Credit Hour: 1.50

Sessional on Object Oriented Programming: Fundamentals of OOP, Classes and Objects, Constructors and Destructors, Inheritance, Polymorphism, Advanced Topics.

EEE 1251

Electronic Devices and Circuits

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Introduction to Semiconductors & Diode: P-N Junction Diode, V-I Characteristics, Light-Emitting Diode (LED), Zener Diode, Diode Applications: Half-Wave and Full-Wave Rectifiers – Operation and Efficiency, Ripple factor.

Linear Wave Shaping: Diode Wave Shaping Techniques, Clipping and Clamping Circuits, Voltage Regulation using Zener Diode.

Bipolar Junction Transistor: NPN and PNP Transistors, Amplifying and Switching Actions of Transistor, Transistor Characteristics and Regions of Operation, CB, CE & CC Configurations, Transistor Load Line and Operating Point, BJT Biasing, Small Signal Equivalent Circuit Models, Small-Signal Analysis of Single-Stage Amplifiers, Designing Logic Gate using BJT.

Field Effect Transistor: Principle of Operation of JFET and MOSFET, Depletion and Enhancement Type NMOS and PMOS, MOSFET as switch and amplifier, CMOS Inverter.

Op-amp: Basic OP-amp Characteristics, Gain, Input and Output Impedance, Feedback, Inverting and Non-Inverting Amplifiers, Integrators, Differentiators, Summing Amplifiers, Introduction to Oscillators, Comparator Circuits, Schmitt Trigger, Linear and Non-Linear Applications of Op-amp.

555 Timer: Architecture of 555 Timer, Monostable, Bistable and Astable Multivibrators using 555 Timer.

Filter Circuits: Filter Fundamentals, Different Types of Filters, Passive Filters, Active Filters.

Power Electronic Devices: SCR, TRIAC, DIAC, UJT Characteristics and Applications; Introduction to IC Fabrication Techniques.

EEE 1252

Electronic Devices and Circuits Sessional

Contact Hours/Week: 3 Hours/Week | Credit Hour: 1.50

Sessional on Electronic Drives and Circuits: Diode, Bipolar Junction Transistor (BJT), Field Effect Transistor (FET), Op-Amps, 555 Timer, Filter Circuit, Logic Gate Design using BJT.

Math 1213

Coordinate Geometry and Ordinary Differential Equation

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Coordinate Geometry of Two Dimensions: Change of Axes, Transformation of Coordinates, Simplification of Equations of Curves.

Coordinate Geometry of Three Dimensions: System of Coordinates, Distance between two Points, Section Formula, Direction Cosines and Projection, Planes and Straight Lines.

Ordinary Differential Equation: Degree and Order of Ordinary Differential Equations. Formation of Differential Equations. Solutions of First Order Differential Equations by Various Methods, Solutions of General Linear Differential Equations of Second and Higher Orders with Constant Coefficients, Solution of Homogeneous Linear Differential Equations. Solution of Higher Order Differential Equations when the Dependent of Independent Variables are Absent. Differential Equations with Variable Coefficients.

Series Solution: Singular Points, Series Solutions: Frobenius Method, Bessel's and Legender's Differential Equations.

Phy 1213

Physics

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Structure of Matter: Structure of Matter. Different Types of Bonds in Solids: Metallic, Van Der Waals', Covalent and Ionic Bond. Packing in Solids: Inter Atomic Distances and Forces of Equilibrium, X-Ray Diffraction, Bragg's Law, Distinction among Insulator, Semiconductor and Conductor.

Atomic Physics: Atom Models: Thomson Atom Model, Rutherford Atom Model, Rutherford Scattering Formula, Electron Orbits, Bohr Atom Model, Energy Levels and Spectra, Particle Properties of Waves: Photoelectric Effect, Einstein's Photoelectric Equation, Laws of Photoelectric Emission, Photovoltaic Cells, Compton Effect, Wave Properties of Particle: De Broglie Waves, Group Velocity, Phase Velocity.

Waves and Oscillations: Oscillations: Simple Harmonic Motion, Composition of Simple Harmonic Motions and Lissajous' Figures, Damped and Forced Oscillations, Resonance. Waves: Traveling and Standing waves, Energy Calculation of Traveling and Standing Waves, Intensity of Waves, Beats, Doppler Effect.

Physical Optics: Theories of Light: Wave theory: Huygen's Wave Theory, Huygen's Principle and Construction, Superposition of Light Waves, Electromagnetic Theory, Particle Theory: Newton's Corpuscular Theory, Quantum Theory of Light.

Interference: Introduction, Conditions of Interference, Young's Double Slit Experiment, Fresnel's Bi-prism, Thin Film Interference, Interference Due to Multiple Reflection, Newton's Ring.

Diffraction: Fresnel and Fraunhofer Diffraction, Diffraction by Single and Double Slit, Diffraction Gratings.

Polarization: Introduction, Methods of Producing Polarized Light, Polarization by Reflection and Refraction, Polarization by Double Refraction, Construction of Nicol Prism. Production and Analysis of Polarized Light, Optical Activity, Optics of Crystals, Polarimeter.

Phy 1214

Physics Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Sessional on Physics: Introduction to Laboratory Works and Experiments, Waves and Oscillations, Atomic and Modern Physics, Physical Optics.